

# 전국기능경기대회 채점기준

## 1. 채점 시 유의사항

직 종 명

클라우드컴퓨팅

※ 다음 사항을 유의하여 채점하시오.

- 1) AWS의 지역은 ap-northeast-2를 사용합니다.
- 2) 웹페이지 접근은 크롬이나 파이어폭스를 이용합니다.
- 3) 웹페이지에서 언어에 따라 문구가 다르게 보일 수 있습니다.
- 4) shell에서의 명령어의 출력은 버전에 따라 조금 다를 수 있습니다.
- 5) 채점은 문항 순서대로 진행해야 합니다.
- 6) 부분 점수가 따로 없는 문항은 모두 맞아야 점수로 인정됩니다.
- 7) 리소스의 정보를 읽어오는 채점항목은 기본적으로 스크립트 결과를 통해 채점을 진행하며, 만약 선수가 이의가 있다면 명령어를 직접 입력하여 확인해볼 수 있습니다.
- 8) (예상 출력)은 바로 이전 (명령어 입력)의 예상 출력을 의미합니다.
- 9) 채점 시에는 별도로 제공한 채점 스크립트(mark.sh)를 실행하여 채점할 수 있습니다. 다만, 선수가 직접 입력을 원할 경우 채점기준표에 명시된 명령어 그대로 입력하여 채점할 수 있습니다.
- 10) 제출 전 mark-sg의 경우 선수 임의로 구성해두도록 합니다.
- 11) 모든 채점 사항은 Cloudshell에 wsc2026-skills-app-sub-a 서브넷과 사전구성한 mark-sg를 연결한 후 진행합니다.
- 12) 11-1 채점을 실행하여 생긴 파드들은 채점이 끝난 후 삭제하도록 합니다.
- 13) 아래 명령어를 사전입력합니다.

```
aws configure set default.region ap-northeast-2
```

```
ACCOUNT_ID=$(aws sts get-caller-identity --query "Account" --output text)
```

```
VPC_ID=$(aws ec2 describe-vpcs --filters "Name=tag:Name,Values=wsc2026-skills-vpc" --query "Vpcs[0].VpcId" --output text)
```

```
BUCKET=$(aws s3api list-buckets --query "Buckets[?contains(Name,'wsc2026-static')].Name" --output text)
```

```
aws eks update-kubeconfig --name wsc2026-eks-cluster --region ap-northeast-2 2>/dev/null
```

```
check_kms() {
```

```
    local ALIAS=$1 ACTUAL_ARN=$2; local EXPECTED_ARN=$(aws kms describe-key --key-id "alias/${ALIAS}" --query "KeyMetadata.Arn" --output text 2>/dev/null); local KEY_ID=$(aws kms describe-key --key-id "alias/${ALIAS}" --query "KeyMetadata.KeyId" --output text 2>/dev/null)
    if [ -z "$KEY_ID" ] || [ "$KEY_ID" = "None" ]; then echo "KMS ${ALIAS}: FAIL (key not found)";
    return; fi
```

```
    if [ "$EXPECTED_ARN" != "$ACTUAL_ARN" ]; then echo "KMS ${ALIAS}: FAIL (wrong key)"; return; fi
```

```
    POLICY=$(aws kms get-key-policy --key-id "$KEY_ID" --policy-name default --output text 2>/dev/null)
```

```
    if echo "$POLICY" | grep -q '"kms:W*"' ; then echo "KMS ${ALIAS}: FAIL (kms:*)"; elif echo "$POLICY" | grep -q ':root'; then echo "KMS ${ALIAS}: FAIL (root)"; else echo "KMS ${ALIAS}: PASS"; fi
```

```
}
```

※ 채점기준 양식은 반드시, 양식에 맞추어 작성해야 합니다.(채점사이트 입력에 필요)

2. 채점기준표

| 1) 주요항목별 배점 |          |                            | 직 종 명 |      | 클라우드컴퓨팅 |           |           |    |
|-------------|----------|----------------------------|-------|------|---------|-----------|-----------|----|
| 과제<br>번호    | 일련<br>번호 | 주요항목                       | 배점    | 채점방법 |         | 채점시기      |           | 비고 |
|             |          |                            |       | 독립   | 합의      | 경기<br>진행중 | 경기<br>종료후 |    |
| 제1과제        | 1        | Networking                 | 2.0   | ○    |         |           | ○         |    |
|             | 2        | Database                   | 1.3   | ○    |         |           | ○         |    |
|             | 3        | Container<br>Registry      | 1.2   | ○    |         |           | ○         |    |
|             | 4        | Container<br>Orchestration | 3.0   | ○    |         |           | ○         |    |
|             | 5        | Deployment                 | 6.5   | ○    |         |           | ○         |    |
|             | 6        | S3                         | 1.0   | ○    |         |           | ○         |    |
|             | 7        | Lambda                     | 3.0   | ○    |         |           | ○         |    |
|             | 8        | ALB                        | 1.3   | ○    |         |           | ○         |    |
|             | 9        | CloudFront                 | 4.0   | ○    |         |           | ○         |    |
|             | 10       | WAF                        | 1.5   | ○    |         |           | ○         |    |
|             | 11       | Observability              | 5.2   | ○    |         |           | ○         |    |
| 합 계         |          |                            | 30    |      |         |           |           |    |

## 2) 채점방법 및 기준

| 과제<br>번호 | 일<br>련<br>번<br>호 | 주요항목                       | 일련<br>번호 | 세부항목(채점방법)              | 배점  |
|----------|------------------|----------------------------|----------|-------------------------|-----|
| 제1과제     | 1                | Networking                 | 1        | VPC                     | 1.0 |
|          |                  |                            | 2        | Routing                 | 1.0 |
|          | 2                | Database                   | 1        | DynamoDB Table          | 1.3 |
|          | 3                | Container Registry         | 1        | ECR Container           | 1.2 |
|          | 4                | Container<br>Orchestration | 1        | EKS Cluster             | 1.5 |
|          |                  |                            | 2        | NodeGroup               | 1.0 |
|          |                  |                            | 3        | EKS IAM                 | 0.5 |
|          | 5                | Deployment                 | 1        | Resource Status         | 1.5 |
|          |                  |                            | 2        | Scheduling & Resources  | 1.0 |
|          |                  |                            | 3        | Probes & ConfigMap      | 1.0 |
|          |                  |                            | 4        | Service & Pod Placement | 1.5 |
|          |                  |                            | 5        | Pod Identity            | 1.5 |
|          | 6                | S3                         | 1        | S3 Bucket               | 1.0 |
|          | 7                | Lambda                     | 1        | Lambda Functions        | 1.5 |
|          |                  |                            | 2        | Lambda IAM              | 1.5 |
|          | 8                | Load Balancer              | 1        | Load Balancer           | 1.3 |
|          | 9                | CloudFront                 | 1        | CloudFront Distribution | 1.2 |
|          |                  |                            | 2        | Cache Policy            | 1.3 |
|          |                  |                            | 3        | POST & GET E2E          | 1.5 |
|          | 10               | WAF                        | 1        | WAF Protection          | 1.5 |
|          | 11               | Observability              | 1        | Observability Deploy    | 1.2 |
|          |                  |                            | 2        | Grafana Datasource      | 1.0 |
|          |                  |                            | 3        | Grafana Dashboard       | 1.5 |
|          |                  |                            | 4        | Alert Firing            | 1.5 |
| 합계       |                  |                            |          |                         | 30  |

| 순번    | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                       |             |                       |             |        |       |           |                       |             |        |       |           |                       |             |        |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------|-----------------------|-------------|--------|-------|-----------|-----------------------|-------------|--------|-------|-----------|-----------------------|-------------|--------|
| 1-1   | <div>1) <pre>aws ec2 describe-vpcs --vpc-ids "\$VPC_ID" --query "Vpcs[0].CidrBlock" --output text;</pre><pre>aws ec2 describe-subnets --filters "Name=vpc-id,Values=\${VPC_ID}" --query "Subnets[0].{Name:Tags[?Key=='Name'].Value [0],Cidr:CidrBlock}" --output text</pre></div> <div>2) VPC와 서브넷의 이름이 정확하게 일치하며 VPC와 서브넷에 지정한 CIDR가 부여되어있는지 확인합니다.</div> <div>192.168.0.0/16</div> <div>192.168.1.0/24 wsc2026-skills-hub-sub-a</div> <div>192.168.10.0/24 wsc2026-skills-hub-sub-b</div> <div>192.168.2.0/24 wsc2026-skills-app-sub-a</div> <div>192.168.20.0/24 wsc2026-skills-app-sub-b</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |             |                       |             |        |       |           |                       |             |        |       |           |                       |             |        |
| 1-2   | <div>1) <pre>IGW_ID=\$(aws ec2 describe-internet-gateways --filters "Name=attachment.vpc-id,Values=\${VPC_ID}" "Name=tag:Name,Values=wsc2026-skills-igw" --query "InternetGateways[0].InternetGatewayId" --output text); echo "IGW: \$IGW_ID"</pre><pre>NAT_A_ID=\$(aws ec2 describe-nat-gateways --filter "Name=vpc-id,Values=\${VPC_ID}" "Name=tag:Name,Values=wsc2026-skills-nat-a" "Name=state,Values=available" --query "NatGateways[0].NatGatewayId" --output text);</pre><pre>NAT_B_ID=\$(aws ec2 describe-nat-gateways --filter "Name=vpc-id,Values=\${VPC_ID}" "Name=tag:Name,Values=wsc2026-skills-nat-b" "Name=state,Values=available" --query "NatGateways[0].NatGatewayId" --output text); echo "NAT-A: \$NAT_A_ID"; echo "NAT-B: \$NAT_B_ID"</pre><pre>aws ec2 describe-route-tables --filters "Name=vpc-id,Values=\${VPC_ID}" "Name=tag:Name,Values=*" --query "RouteTables[0].{Name:Tags[?Key=='Name'].Value [0],Route:Routes[?DestinationCidrBlock=='0.0.0.0/0'] [0]}" --output text</pre></div> <div>2) 실행 결과가 강조된 텍스트와 동일하며, hub-rtb에는 igw의 ID가, app-rtb-a에는 NAT-A의 ID가, app-rtb-b에는 NAT-B의 ID가 서로 일치하게 연결되어 있는지 확인합니다.</div> <div>IGW: igw-0a85cd67732857621</div> <div>NAT-A: nat-071e3d4820939525d</div> <div>NAT-B: nat-03f60025e555f65bb</div> <div>wsc2026-skills-app-rtb-b</div> <div><table><tr><td>ROUTE</td><td>0.0.0.0/0</td><td>nat-03f60025e555f65bb</td><td>CreateRoute</td><td>active</td></tr></table></div> <div>wsc2026-skills-hub-rtb</div> <div><table><tr><td>ROUTE</td><td>0.0.0.0/0</td><td>igw-0a85cd67732857621</td><td>CreateRoute</td><td>active</td></tr></table></div> <div>wsc2026-skills-app-rtb-a</div> <div><table><tr><td>ROUTE</td><td>0.0.0.0/0</td><td>nat-071e3d4820939525d</td><td>CreateRoute</td><td>active</td></tr></table></div> <div>순서가 다를 수 있으나 app-rtb 들의 경우 nat-*를 사용하고, hub-rtb들의 경우 igw를 사용해야 합니다.</div> | ROUTE                 | 0.0.0.0/0   | nat-03f60025e555f65bb | CreateRoute | active | ROUTE | 0.0.0.0/0 | igw-0a85cd67732857621 | CreateRoute | active | ROUTE | 0.0.0.0/0 | nat-071e3d4820939525d | CreateRoute | active |
| ROUTE | 0.0.0.0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | nat-03f60025e555f65bb | CreateRoute | active                |             |        |       |           |                       |             |        |       |           |                       |             |        |
| ROUTE | 0.0.0.0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | igw-0a85cd67732857621 | CreateRoute | active                |             |        |       |           |                       |             |        |       |           |                       |             |        |
| ROUTE | 0.0.0.0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | nat-071e3d4820939525d | CreateRoute | active                |             |        |       |           |                       |             |        |       |           |                       |             |        |

| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-1 | <p>1) aws dynamodb describe-table --table-name wsc2026-book-table --query "Table.[KeySchema[0].AttributeName,BillingModeSummary.BillingMode,SSEDescription.SSEType,DeletionProtectionEnabled,GlobalSecondaryIndexes[0].KeySchema[0].AttributeName]" --output text 2&gt;/dev/null</p> <p>aws dynamodb describe-continuous-backups --table-name wsc2026-book-table --query "ContinuousBackupsDescription.[PointInTimeRecoveryDescription.PointInTimeRecoveryStatus,PointInTimeRecoveryDescription.RecoveryPeriodInDays,ContinuousBackupsStatus]" --output text 2&gt;/dev/null</p> <p>aws dynamodb get-resource-policy --resource-arn "arn:aws:dynamodb:ap-northeast-2:\${ACCOUNT_ID}:table/wsc2026-book-table" --query "Policy" --output text 2&gt;/dev/null   python3 -c "import sys,json:[print(f'{s.get(W\"ActionW\",W\"W\")} : {s.get(W\"PrincipalW\",{}).get(W\"AWSW\",W\"W\").split(W\"/W\")[1]}') for s in json.loads(sys.stdin.read()).get('Statement',[])]" 2&gt;/dev/null    echo "No resource policy"</p> <p>check_kms "wsc2026-db-kms" "\$(aws dynamodb describe-table --table-name wsc2026-book-table --query 'Table.SSEDescription.KMSMasterKeyArn' --output text 2&gt;/dev/null)"</p> <p>2) 실행 결과가 아래와 정확히 일치해야 합니다. 단, 강조된 부분의 출력 순서는 무관합니다.</p> <pre>client_id      PAY_PER_REQUEST KMS      True      booking_id ENABLED 35      ENABLED</pre> <p><b><u>dynamodb:PutItem : wsc2026-book-pod-role</u></b><br/> <b><u>dynamodb:Query : wsc2026-book-function-role</u></b></p> <p>KMS wsc2026-db-kms: PASS</p> |
| 3-1 | <p>1) aws ecr describe-repositories --repository-names wsc2026-book-ecr --query "repositories[0].[imageScanningConfiguration.scanOnPush,imageTagMutability,imageTagMutabilityExclusionFilters[0].filter,encryptionConfiguration.encryptionType]" --output text 2&gt;/dev/null; aws ecr list-images --repository-name wsc2026-book-ecr --query "imageIds[].imageTag" --output text 2&gt;/dev/null</p> <p>check_kms "wsc2026-ecr-kms" "\$(aws ecr describe-repositories --repository-names wsc2026-book-ecr --query 'repositories[0].encryptionConfiguration.kmsKey' --output text 2&gt;/dev/null)"</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <pre>True      MUTABLE_WITH_EXCLUSION v1*      KMS v1.0.0</pre> <p>KMS wsc2026-ecr-kms: PASS</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4-1 | <pre> 1) aws eks describe-cluster --name wsc2026-eks-cluster --query "cluster.[version,status,resourcesVpcConfig.endpointPublicAccess,resourcesVpcConfig.endpointPrivateAccess,logging.clusterLogging[0].enabled]" --output text 2&gt;/dev/null CLUSTER_SG=\$(aws eks describe-cluster --name wsc2026-eks-cluster --query "cluster.resourceVpcConfig.clusterSecurityGroupId" --output text 2&gt;/dev/null); ANYIP=\$(aws ec2 describe-security-groups --group-ids "\$CLUSTER_SG" --query "SecurityGroups[0].IpPermissions[?IpProtocol=='-1'].IpRanges[].CidrIp" --output text 2&gt;/dev/null); if [ -n "\$ANYIP" ]; then echo "SG: FAIL"; else echo "SG: PASS"; fi kubectI get configmap coredns -n kube-system -o jsonpath='{.data.Corefile}' 2&gt;/dev/null   grep -oP 'kubernetes WK[^ ]+' check_kms "wsc2026-eks-kms" "\$(aws eks describe-cluster --name wsc2026-eks-cluster --query 'cluster.encryptedConfig[0].provider.keyArn' --output text 2&gt;/dev/null)" 2) 실행 결과가 아래와 동일한지 확인합니다. 1.35    ACTIVE    False    True    True SG: PASS wsc2026.skills.local KMS wsc2026-eks-kms: PASS </pre>                                                                                         |
| 4-2 | <pre> 1) NG1=\$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-addon-nodegroup --query "nodegroup.[nodegroupName,instanceTypes[0]]" --output text 2&gt;/dev/null); echo "\$NG1          wsc2026/node=addon" ADDON_COUNT=\$(kubectI get nodes -l wsc2026/node=addon --no-headers --request-timeout=10s 2&gt;/dev/null   wc -l) if [ "\$ADDON_COUNT" -ge 1 ]; then echo "Addon Nodes: PASS (\$ADDON_COUNT)"; else echo "Addon Nodes: FAIL"; fi NG2=\$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-workload-ng --query "nodegroup.[nodegroupName,instanceTypes[0]]" --output text 2&gt;/dev/null); echo "\$NG2          wsc2026/node=application" WORK_COUNT=\$(kubectI get nodes -l wsc2026/node=application --no-headers --request-timeout=10s 2&gt;/dev/null   wc -l) if [ "\$WORK_COUNT" -ge 1 ]; then echo "Workload Nodes: PASS (\$WORK_COUNT)"; else echo "Workload Nodes: FAIL"; fi 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-addon-nodegroup t3.medium          wsc2026/node=addon Addon Nodes: PASS (2) wsc2026-workload-ng      t3.medium          wsc2026/node=application Workload Nodes: PASS (2) </pre> |

| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4-3 | <p>1) CR=\$(aws eks describe-cluster --name wsc2026-eks-cluster --query "cluster.roleArn" --output text 2&gt;/dev/null   awk -F/ '{print \$NF}'); AR=\$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-addon-nodegroup --query "nodegroup.nodeRole" --output text 2&gt;/dev/null   awk -F/ '{print \$NF}'); WR=\$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-workload-ng --query "nodegroup.nodeRole" --output text 2&gt;/dev/null   awk -F/ '{print \$NF}')</p> <p>for PAIR in "Cluster Role:\$CR" "Addon Node Role:\$AR" "Workload Node Role:\$WR"; do LABEL="\${PAIR%:*}"; ROLE="\${PAIR#*:}"; ADMIN=\$(aws iam list-attached-role-policies --role-name "\$ROLE" --query "AttachedPolicies[?PolicyName=='AdministratorAccess'].PolicyName" --output text 2&gt;/dev/null); if [ -n "\$ADMIN" ]; then echo "\$LABEL: FAIL"; else echo "\$LABEL: PASS"; fi; done</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>Cluster Role: PASS</p> <p>Addon Node Role: PASS</p> <p>Workload Node Role: PASS</p> |
| 5-1 | <p>1) kubectl get deploy,svc,ingress,pdb -n wsc2026 --no-headers 2&gt;/dev/null   awk ' /^deployment/ { split(\$2,a,"/"); print "Deployment: " (a[1]==a[2] &amp;&amp; a[1]=="2" ? "PASS" : "FAIL") } /^service/ { print "Service: PASS" } /^ingress/ { print "Ingress: " (\$4 ~ /W.elbW.amazonawsW.com/ ? "PASS (" \$4 )" : "FAIL") } /^poddisrupt/ { print "PDB: " (\$2=="1" ? "PASS" : "FAIL") } ,</p> <p>2) 실행결과가 아래와 동일한지 확인하며, Ingress의 ALB DNS는 wsc2026-app-alb가 포함되었는지 확인합니다.</p> <p>Deployment: PASS</p> <p>Service: PASS</p> <p>Ingress: PASS (wsc2026-app-alb-<a href="#">random</a>.ap-northeast-2.elb.amazonaws.com)</p> <p>PDB: PASS</p>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5-2 | <p>1) kubectl get deploy wsc2026-book-deploy -n wsc2026 -o jsonpath='{.spec.replicas:.spec.replicas} node:{.spec.template.spec.nodeSelector.wsc2026/node} topo:{.spec.template.spec.topologySpreadConstraints[0].topologyKey} cpu:{.spec.template.spec.containers[0].resources.requests.cpu} mem:{.spec.template.spec.containers[0].resources.requests.memory}' 2&gt;/dev/null</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>replicas:2 node:application topo:topology.kubernetes.io/zone cpu:250m mem:512Mi</p> <p><a href="#">3번째 topology의 경우 zone, subnet 과 같은 단어가 포함되어 가용영역 분산을 진행했으면 인정</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5-3 | <p>1) <code>kubectl get deploy wsc2026-book-deploy -n wsc2026 -o jsonpath='{.spec.template.spec.containers[0].startupProbe.httpGet.path}:{.spec.template.spec.containers[0].startupProbe.httpGet.port} readiness:{.spec.template.spec.containers[0].readinessProbe.httpGet.path}:{.spec.template.spec.containers[0].readinessProbe.httpGet.port} liveness:{.spec.template.spec.containers[0].livenessProbe.httpGet.path}:{.spec.template.spec.containers[0].livenessProbe.httpGet.port}' 2&gt;/dev/null; echo; kubectl get configmap book-config -n wsc2026 -o jsonpath='{.data}' 2&gt;/dev/null; echo</code></p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>startup:/health:8080 readiness:/health:8080 liveness:/health:8080<br/>{"AWS_REGION":"ap-northeast-2","TABLE_NAME":"wsc2026-book-table"}</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5-4 | <p>1) <code>WORKLOAD_NODES=\$(kubectl get nodes -l wsc2026/node=application --no-headers -o custom-columns='NAME:.metadata.name' 2&gt;/dev/null)</code><br/> <code>kubectl get pods -n wsc2026 -o custom-columns='NAME:.metadata.name,STATUS:.status.phase,NODE:.spec.nodeName' --no-headers 2&gt;/dev/null   while read NAME STATUS NODE; do</code><br/> <code>echo "\$NAME: \$(echo "\$WORKLOAD_NODES"   grep -q "^\$NODE\$" &amp;&amp; echo PASS    echo FAIL)"</code><br/> <code>done</code></p> <p>2) 실행 결과에서 wsc2026-book-deploy가 포함된 텍스트가 2개 출력되며, 모두 PASS인지 확인합니다.</p> <p>wsc2026-book-deploy-7b88597d86-d6l72: PASS<br/>wsc2026-book-deploy-7b88597d86-z8nkt: PASS</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5-5 | <p>1) <code>PI_SA=\$(aws eks list-pod-identity-associations --cluster-name wsc2026-eks-cluster --namespace wsc2026 --query 'associations[0].serviceAccount' --output text 2&gt;/dev/null)</code><br/> <code>PI_ROLE=\$(aws eks list-pod-identity-associations --cluster-name wsc2026-eks-cluster --namespace wsc2026 --query 'associations[0].associationId' --output text 2&gt;/dev/null)</code><br/> <code>[ "\$PI_SA" = "wsc2026-book-sa" ] &amp;&amp; echo "Pod Identity SA: PASS (\$PI_SA)"    echo "Pod Identity SA: FAIL (\$PI_SA)"</code><br/> <code>ROLE_NAME=\$(aws eks describe-pod-identity-association --cluster-name wsc2026-eks-cluster --association-id "\$PI_ROLE" --query 'association.roleArn' --output text 2&gt;/dev/null awk -F/ '{print \$NF}')</code><br/> <code>MANAGED_ACTIONS=\$(aws iam list-attached-role-policies --role-name "\$ROLE_NAME" --query 'AttachedPolicies[].PolicyArn' --output text 2&gt;/dev/null tr 'Wt' 'Wn' while read arn;do</code><br/> <code>VERSION=\$(aws iam get-policy --policy-arn "\$arn" --query 'Policy.DefaultVersionId' --output text 2&gt;/dev/null);aws iam get-policy-version --policy-arn "\$arn" --version-id "\$VERSION" --query 'PolicyVersion.Document.Statement[].Action' --output text</code><br/> <code>2&gt;/dev/null;done)</code><br/> <code>echo "\$MANAGED_ACTIONS" grep -q dynamodb:PutItem &amp;&amp; ! echo "\$MANAGED_ACTIONS" grep -qE '^W*\$' &amp;&amp; echo "Pod Identity Role: PASS"    echo "Pod Identity Role: FAIL"</code></p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>Pod Identity SA: PASS (wsc2026-book-sa)<br/>Pod Identity Role: PASS</p> |



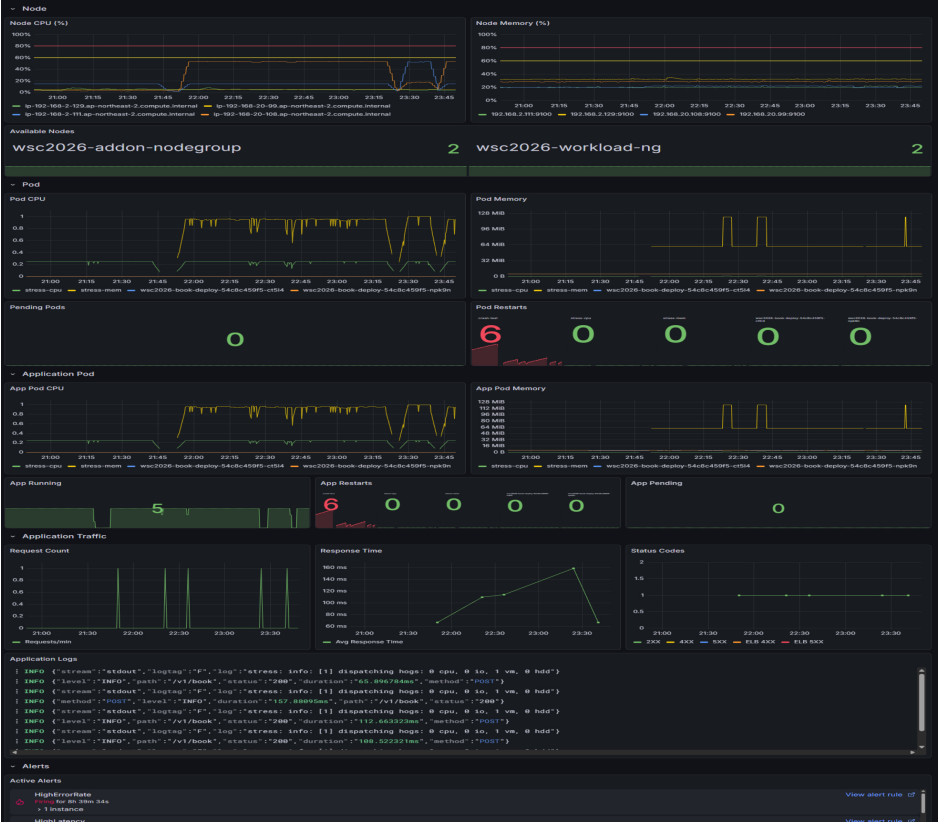
| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| 6-1 | <pre> 1) echo "\$BUCKET"; aws s3api get-public-access-block --bucket "\$BUCKET" --query "PublicAccessBlockConfiguration.[BlockPublicAcls,BlockPublicPolicy,IgnorePublicAcls,RestrictPublicBuckets]" --output text 2&gt;/dev/null; aws s3api get-bucket-encryption --bucket "\$BUCKET" --query "ServerSideEncryptionConfiguration.Rules[0].{SSE:ApplyServerSideEncryptionByDefault.SSEAlgorithm,BucketKey:BucketKeyEnabled}" --output text 2&gt;/dev/null; aws s3api list-objects --bucket "\$BUCKET" --prefix "static/" --query "Contents[?Size&gt;W`0W`.Key]" --output text 2&gt;/dev/null check_kms "wsc2026-bucket-kms" "\$(aws s3api get-bucket-encryption --bucket "\$BUCKET" --query 'ServerSideEncryptionConfiguration.Rules[0].ApplyServerSideEncryptionByDefault.KMSMasterKeyID' --output text 2&gt;/dev/null)" BUCKET_KMS_ARN=\$(aws kms describe-key --key-id "alias/wsc2026-bucket-kms" --query "KeyMetadata.Arn" --output text 2&gt;/dev/null) echo "S3 Object KMS Check:"; for OBJ in \$(aws s3api list-objects --bucket "\$BUCKET" --prefix "static/" --query "Contents[].Key" --output text 2&gt;/dev/null); do     KEY_ID=\$(aws s3api head-object --bucket "\$BUCKET" --key "\$OBJ" --query "SSEKMSKeyId" --output text 2&gt;/dev/null)     if [ "\$KEY_ID" = "\$BUCKET_KMS_ARN" ]; then echo " \$OBJ: PASS"; else echo " \$OBJ: FAIL (\$KEY_ID)"; fi done 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-static-&lt;임의 영문 4자리&gt;-&lt;선수 비번호&gt;-bucket True    True    True    True True    aws:kms static/index.html      static/main.jpeg KMS wsc2026-bucket-kms: PASS S3 Object KMS Check: static/index.html: PASS static/main.jpeg: PASS </pre> |
| 7-1 | <pre> 1) aws lambda get-function --function-name wsc2026-book-get-function --query "Configuration.{Name:FunctionName,Runtime:Runtime}" --output json 2&gt;/dev/null aws lambda get-function --function-name wsc2026-book-get-function --query "Configuration.Environment.Variables" --output json 2&gt;/dev/null check_kms "wsc2026-function-kms" "\$(aws lambda get-function --function-name wsc2026-book-get-function --query 'Configuration.KMSKeyArn' --output text 2&gt;/dev/null)" 2) 실행 결과가 아래와 동일한지 확인하며, TABLE_NAME은 평문이 아닌 KMS로 암호화된 텍스트가 출력되는지 확인합니다. {   "Name": "wsc2026-book-get-function",   "Runtime": "python3.12" } {   "TABLE_NAME": "AQICAHjHYzx8HKYIOX8fnCJbPqPanGMX7gxnAX57CFiSG1wT1wEpFZotnxm3...." } KMS wsc2026-function-kms: PASS </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| 순번  | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 7-2 | <p>1) <code>LAMBDA_ROLE=\$(aws lambda get-function --function-name wsc2026-book-get-function --query "Configuration.Role" --output text 2&gt;/dev/null   awk -F/ '{print \$NF}')</code>; echo "\$LAMBDA_ROLE"; <code>POLICIES=\$(aws iam list-attached-role-policies --role-name "\$LAMBDA_ROLE" --query "AttachedPolicies[].PolicyName" --output text 2&gt;/dev/null)</code>; echo "\$POLICIES"; if echo "\$POLICIES"   grep -q "AdministratorAccess"; then echo "Role: FAIL (Admin)"; else echo "Role: PASS"; fi</p> <p><code>POLICY_ARN=\$(aws iam list-attached-role-policies --role-name "\$LAMBDA_ROLE" --query "AttachedPolicies[?PolicyName!='AWSLambdaBasicExecutionRole'].PolicyArn[0]" --output text 2&gt;/dev/null)</code>; <code>ACTIONS=\$(aws iam get-policy-version --policy-arn "\$POLICY_ARN" --version-id \$(aws iam get-policy --policy-arn "\$POLICY_ARN" --query "Policy.DefaultVersionId" --output text 2&gt;/dev/null) --query "PolicyVersion.Document.Statement[].Action" --output text 2&gt;/dev/null)</code></p> <p>if echo "\$ACTIONS"   grep -q "dynamodb:Query" &amp;&amp; ! echo "\$ACTIONS"   grep -q 'W*'; then echo "Policy: PASS"; else echo "Policy: FAIL"; fi</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>wsc2026-book-function-role<br/>wsc2026-book-function-policy (AWSLambdaBasicExecutionRole) &lt;-해당 이름가진 정책 출력가능<br/>Role: PASS<br/>Policy: PASS</p> |
| 8-1 | <p>1) <code>aws elbv2 describe-load-balancers --names wsc2026-app-alb --query "LoadBalancers[0].Scheme" --output text 2&gt;/dev/null</code>; <code>ALB_SGS=\$(aws elbv2 describe-load-balancers --names wsc2026-app-alb --query "LoadBalancers[0].SecurityGroups[]" --output text 2&gt;/dev/null)</code>; <code>aws ec2 describe-security-groups --group-ids \$ALB_SGS --query "SecurityGroups[].GroupName" --output text 2&gt;/dev/null</code></p> <p><code>ALB_DNS=\$(aws elbv2 describe-load-balancers --names wsc2026-app-alb --query "LoadBalancers[0].DNSName" --output text 2&gt;/dev/null)</code>; <code>ALB_CODE=\$(curl -s -o /dev/null -w "%{http_code}" --max-time 5 "http://\${ALB_DNS}/" 2&gt;/dev/null)</code>; if [ "\$ALB_CODE" = "000" ]; then echo "ALB direct: BLOCKED"; else echo "ALB direct: FAIL (\$ALB_CODE)"; fi</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>internet-facing<br/>wsc2026-app-alb-sg<br/>ALB direct: BLOCKED</p>                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 9-1 | <p>1) <code>CF_ARN=\$(aws resourcegroupstaggingapi get-resources --tag-filters Key=Name,Values=wsc2026-cdn --resource-type-filters cloudfront --region us-east-1 --query "ResourceTagMappingList[0].ResourceARN" --output text 2&gt;/dev/null)</code>; <code>CF_ID=\$(echo "\$CF_ARN"   awk -F/ '{print \$NF}')</code>; <code>CF_DOMAIN=\$(aws cloudfront get-distribution --id "\$CF_ID" --region us-east-1 --query "Distribution.DomainName" --output text 2&gt;/dev/null)</code>; <code>curl -s -o /dev/null -w "\${CF_DOMAIN} : %{http_code}" "https://\${CF_DOMAIN}/" 2&gt;/dev/null</code>; echo</p> <p>2) 실행 결과에서 Cloudfront 도메인과 200이 출력되는지 확인합니다.</p> <p>&lt;random&gt;.cloudfront.net : 200</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| 순번   | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| 9-2  | <p>1) DISABLED="4135ea2d-6df8-44a3-9df3-4b5a84be39ad"; OPTIMIZED="658327ea-f89d-4fab-a63d-7e88639e58f6"; DEFAULT_CP=\$(aws cloudfront get-distribution --id "\$CF_ID" --region us-east-1 --query "Distribution.DistributionConfig.DefaultCacheBehavior.CachePolicyId" --output text 2&gt;/dev/null); if [ "\$DEFAULT_CP" = "\$OPTIMIZED" ]; then echo "S3: CachingOptimized"; else echo "S3: FAIL"; fi</p> <p>for B in \$(aws cloudfront get-distribution --id "\$CF_ID" --region us-east-1 --query "Distribution.DistributionConfig.CacheBehaviors.Items[].PathPattern" --output text 2&gt;/dev/null); do CP=\$(aws cloudfront get-distribution --id "\$CF_ID" --region us-east-1 --query "Distribution.DistributionConfig.CacheBehaviors.Items[?PathPattern=='\${B}'].CachePolicyId[0]" --output text 2&gt;/dev/null); if [ "\$CP" = "\$DISABLED" ]; then echo "ALB/Lambda: CachingDisabled"; break; fi; done</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>S3: CachingOptimized</p> <p>ALB/Lambda: CachingDisabled</p>                                                                                                                                                                                                                                                                                                                                                                      |
| 9-3  | <p>1) BOOKING_ID=\$(curl -s -X POST "https://\${CF_DOMAIN}/booking" -H "Content-Type: application/json" -d '{"client_id":"MARK001","username":"Marker","email":"mark@test.com","concert_name":"TestConcert"}' 2&gt;/dev/null   python3 -c "import sys,json;print(json.load(sys.stdin).get('booking_id',''))" 2&gt;/dev/null); echo "POST booking_id: \$BOOKING_ID"</p> <p>curl -s "https://\${CF_DOMAIN}/v1/book?booking_id=\${BOOKING_ID}" 2&gt;/dev/null; echo</p> <p>2) 필드 순서(client_id, username, email, concert_name, created_at)가 차례대로 일치하는지, 그리고 created_at 시간이 스크립트 실행 시간 기준 1분 이내인지 확인합니다.</p> <p>POST booking_id: &lt;Random&gt;</p> <p>{"client_id": "MARK001", "username": "Marker", "email": "mark@test.com", "concert_name": "TestConcert", "created_at": &lt;실행기준 1분 이내&gt;}</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 10-1 | <p>1) WAF_NAME=\$(aws wafv2 list-web-acls --scope CLOUDFRONT --region us-east-1 --query "WebACLs[?contains(Name, 'wsc2026')].Name" --output text 2&gt;/dev/null)</p> <p>echo "WAF Name: \$WAF_NAME"</p> <p>SQLI=\$(curl -s -o /dev/null -w '%{http_code}' --max-time 5 "https://\${CF_DOMAIN}/v1/book?booking_id=1'%200R%201=1--"); echo "SQLi: \$SQLI"</p> <p>XSS=\$(curl -s -o /dev/null -w '%{http_code}' --max-time 5 "https://\${CF_DOMAIN}/v1/book?booking_id=&lt;script&gt;alert(1)&lt;/script&gt;"); echo "XSS: \$XSS"</p> <p>(for i in \$(seq 1 15); do for j in \$(seq 1 25); do curl -s -o /dev/null --max-time 3 "https://\${CF_DOMAIN}/" &amp; done; sleep 0.4; done; wait) &gt;/dev/null 2&gt;&amp;1; WAF_ID=\$(aws wafv2 list-web-acls --scope CLOUDFRONT --region us-east-1 --query "WebACLs[?contains(Name, 'wsc2026')].Id" --output text 2&gt;/dev/null); RATE_LIMIT=\$(aws wafv2 get-web-acl --scope CLOUDFRONT --region us-east-1 --name "\$WAF_NAME" --id "\$WAF_ID" --query "WebACL.Rules[?Statement.RateBasedStatement].Statement.RateBasedStatement.Limit" --output text 2&gt;/dev/null); if [ -n "\$RATE_LIMIT" ] &amp;&amp; [ "\$RATE_LIMIT" -le 200 ] 2&gt;/dev/null; then echo "Rate: PASS (403)"; else echo "Rate: FAIL (-)"; fi</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>WAF Name: wsc2026-waf</p> <p>SQLi: 403</p> <p>XSS: 403</p> <p>Rate: PASS (403)</p> |

| 순번   | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 11-1 | <p>1) for p in fluent-bit prometheus grafana; do kubectl get pods -n observability --no-headers --request-timeout=10s 2&gt;/dev/null   grep -c "\$p.*Running"   xargs -l{} echo "\$p: {}"; done</p> <p>2) 실행 결과가 1 이상인지 확인합니다.</p> <p>fluent-bit: 4</p> <p>prometheus: 7</p> <p>grafana: 1</p> <p><a href="#">해당 네임스페이스에 모니터링 관련 Pod가 하나라도 있으면 득점.</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 11-2 | <p>1) echo "Datasources:"; curl -s -u admin:'Skills\$\$\$@!' "http://\${GRAFANA_LB}/api/datasources" 2&gt;/dev/null   python3 -c "import sys,json:[print(f' {d[W"nameW"]} ({d[W"typeW"]})') for d in json.load(sys.stdin)]" 2&gt;/dev/null    echo " Grafana unreachable"</p> <p>echo "Dashboards:"; curl -s -u admin:'Skills\$\$\$@!' "http://\${GRAFANA_LB}/api/search?query=wsc2026" 2&gt;/dev/null   python3 -c "import sys,json:[print(f' {d[W"titleW"]}') for d in json.load(sys.stdin)]" 2&gt;/dev/null    echo " Not found"</p> <p>2) 실행 결과가 아래와 동일한지 확인합니다.</p> <p>Datasources:</p> <ul style="list-style-type: none"> <li>alertmanager (alertmanager)</li> <li>cloudwatch (cloudwatch)</li> <li>prometheus (prometheus)</li> </ul> <p>Dashboards:</p> <ul style="list-style-type: none"> <li>wsc2026-grafana-dashboard</li> </ul> |

| 순번   | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 11-3 | <p>1) 아래 명령을 수동으로 입력합니다.</p> <pre> SVC_IP=\$(kubectl get svc -n wsc2026 -o jsonpath='{.items[0].spec.clusterIP}' 2&gt;/dev/null); kubectl r un not-ready --image=busybox --restart=Always -n wsc2026 --overrides='{ "spec": { "tolerations": [ { "oper ator": "Exists" } ], "nodeSelector": { "wsc2026/node": "application" }, "containers": [ { "name": "not-ready", "im age": "busybox", "readinessProbe": { "httpGet": { "path": "/health", "port": 80 }, "periodSeconds": 3 }, "command ": [ "sh", "-c", "sleep 3600" ] ] } } }' &amp;&gt;/dev/null; kubectl run error-gen --image=curlimages/curl --restart =Never -n wsc2026 --overrides='{ "spec": { "tolerations": [ { "operator": "Exists" } ], "nodeSelector": { "wsc20 26/node": "application" } } }' -- sh -c "while true; do curl -s -o /dev/null http://'\$SVC_IP'/nonexis t; sleep 0.1; done" &amp;&gt;/dev/null; kubectl run latency-gen --image=curlimages/curl --restart=Never -n wsc2026 --overrides='{ "spec": { "tolerations": [ { "operator": "Exists" } ], "nodeSelector": { "wsc2026/node": " application" } } }' -- sh -c "while true; do curl -s -o /dev/null http://'\$SVC_IP'/delay?ms=5000; sle ep 0.2; done" &amp;&gt;/dev/null  kubectl run crash-test --image=busybox --restart=Always -n wsc2026 --overrides='{ "spec": { "toleration s": [ { "operator": "Exists" } ], "nodeSelector": { "wsc2026/node": "application" } } }' -- sh -c 'exit 1' &amp;&gt;/dev /null; kubectl run stress-cpu --image=busybox --restart=Never -n wsc2026 --overrides='{ "spec": { "tole rations": [ { "operator": "Exists" } ], "nodeSelector": { "wsc2026/node": "application" }, "containers": [ { "name ": "stress-cpu", "image": "busybox", "resources": { "requests": { "cpu": "250m" }, "limits": { "cpu": "250m" } }, "co mmand": [ "sh", "-c", "while true; do :: done" ] ] } } }' &amp;&gt;/dev/null; kubectl run stress-mem --image=polinux /stress --restart=Never -n wsc2026 --overrides='{ "spec": { "tolerations": [ { "operator": "Exists" } ], "node Selector": { "wsc2026/node": "application" }, "containers": [ { "name": "stress-mem", "image": "polinux/stress ", "resources": { "requests": { "memory": "64Mi" }, "limits": { "memory": "64Mi" } }, "command": [ "stress", "--vm", " 1", "--vm-bytes", "60M", "--vm-keep", "-t", "3600" ] ] } } }' &amp;&gt;/dev/null  sleep 180  GRAFANA_LB=\$(kubectl get svc -n observability -o jsonpath='{range .items[?(@.spec.type=="LoadBalance r")]}{.status.loadBalancer.ingress[0].hostname}{end}' 2&gt;/dev/null) </pre> <p>... 다음 페이지 이어서</p> |

| 순번   | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| 11-3 | <p>2) 출력된 Grafana URL을 통해 admin/Skills\$#\$@!에 로그인 후 wsc2026-grafana-dashboard에서 , 모든 메트릭 및 로그는 빈값이 없어야합니다. 다음 구성 요소가 대쉬보드에 포함되어 있는지 확인합니다.</p> <ul style="list-style-type: none"> <li>- Node CPU (%) - 노드별 CPU 사용률</li> <li>- Node Memory (%) - 노드별 Memory 사용률</li> <li>- Application Logs - 어플리케이션 로그</li> <li>- Application Logs 패널 로그 형식 예시 (/v1/book을 제외한 로그가 있을 경우 응답처리):<br/>INFO {"level":"INFO","path":"/v1/book","status":"200","duration":"112.663323ms","method":"POST"}</li> </ul> <p>(참고 사진, 해당 사진에서 Node CPU, Node MEM, App log만 확인합니다.)</p>  |

| 순번   | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11-4 | <p>1) Active Alerts 패널에 Edit을 선택해 나온 결과와 채점지 사진과 일치하는지 확인합니다.<br/>(* HighLatency Alert 구성요소는 채점과 무관합니다.)</p> <div><div>Active Alerts</div><div><div><div>HighErrorRate</div><div><div></div><div>Firing for 2h 14m 57s</div><div>&gt; 1 instance</div></div><div>View alert rule </div></div><div><div>HighLatency</div><div><div></div><div>Firing for 2h 14m 57s</div><div>&gt; 1 instance</div></div><div>View alert rule </div></div><div><div>PodCrashLooping</div><div><div></div><div>Firing for 4m 49s</div><div>&gt; 1 instance</div></div><div>View alert rule </div></div><div><div>PodHighCPU</div><div><div></div><div>Firing for 2m 49s</div><div>&gt; 1 instance</div></div><div>View alert rule </div></div><div><div>PodHighMemory</div><div><div></div><div>Firing for 6m 49s</div><div>&gt; 1 instance</div></div><div>View alert rule </div></div><div><div>PodNotReady</div><div><div></div><div>Firing for 6m 49s</div><div>&gt; 1 instance</div></div><div>View alert rule </div></div></div></div> |